

Segmented Campus Network Design and Simulation using Cisco Packet Tracer

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Objectives:

1. to make all switches and routers should be protected by password.
2. to design One of the departments would use the DHCP protocol to assign the IP addresses to all hosts of that network.
3. to apply VLAN system, on one of the departments that would separate students and faculties networks.
4. to use a web server and a DNS server should be placed under the admin office.
5. to apply any of our known dynamic routing protocols should be applied in the designed topology.
6. to apply our desired ACL in routers to access the resources of the admin office. (For example, permit/deny any host/network to access the web server or any host in the admin office).

Introduction:

This project focuses on designing and configuring a campus network using Cisco Packet Tracer, integrating multiple networking concepts such as VLAN, DHCP, RIP routing, ACLs, password protection, and server configuration. The network consists of multiple departments—each designed with specific purposes and configurations—to simulate a realistic organizational environment.

The Admin Office manages central services like the Web Server and DNS Server, which are accessible by different departments based on access policies. Department A uses the DHCP protocol to dynamically assign IP addresses to hosts, while Department B uses VLANs to separate faculty and student networks. The routers are configured with RIP (Routing Information Protocol) for inter-department communication, and Access Control Lists (ACLs) are implemented to restrict or allow access to particular resources.

Security has been ensured through password protection on all routers and switches, preventing unauthorized configuration access. This project demonstrates the complete lifecycle of campus network deployment from topology design and logical IP planning to implementation, routing, and access management. By the end, the network achieves full connectivity between authorized hosts and controlled access to the central administrative servers, fulfilling all the requirements of a secure, scalable, and efficient enterprise network.

Theory:

The project is based on the fundamental concepts of computer networking and protocols used to design and manage a campus network efficiently. Each configuration component plays a specific role in ensuring secure, scalable, and organized data communication.

Password protection ensures device security by restricting unauthorized access to router and switch configuration modes. DHCP (Dynamic Host Configuration Protocol) automatically assigns IP

addresses and network parameters to hosts, reducing manual errors and administrative effort. VLAN (Virtual Local Area Network) separates users logically within the same physical network, improving performance, manageability, and security by controlling broadcast domains.

Password Protection:

Password protection is applied to secure router and switch access. It prevents unauthorized users from entering privileged or configuration modes.

Example Configuration:

```
Router(config)# enable secret admin123
Router(config)# line console 0
Router(config-line)# password cisco
Router(config-line)# login
Router(config)# line vty 0 4
Router(config-line)# password remote
Router(config-line)# login
Router(config)# service password-encryption
```

DHCP Protocol:

The Dynamic Host Configuration Protocol (DHCP) automatically assigns IP addresses, subnet masks, and gateways to hosts. It reduces manual configuration errors and simplifies management.

Example Configuration:

```
Router(config)# ip dhcp pool DEPTA_POOL
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 192.168.10.2
```

VLAN Protocol:

VLANs (Virtual Local Area Networks) logically divide a network into multiple broadcast domains on the same switch. This improves security and reduces congestion. Each VLAN is assigned an ID and subnet.

Example Configuration:

```
Switch(config)# vlan 10
Switch(config-vlan)# name STUDENTS
Switch(config)# vlan 20
Switch(config-vlan)# name FACULTY
Switch(config)# interface range f0/1-10
Switch(config-if-range)# switchport access vlan 10
```

```
Switch(config)# interface range f0/11-20
Switch(config-if-range)# switchport access vlan 20
```

ACL Protocol:

Access Control Lists (ACLs) control which hosts or networks can access specific services or devices. They enhance security by permitting or denying traffic based on conditions.

Example Configuration:

```
Router(config)# access-list 110 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config)# access-list 110 permit ip any any
Router(config)# interface g0/0
Router(config-if)# ip access-group 110 in
```

RIP (Routing Information Protocol):

RIP is a dynamic routing protocol that allows routers to exchange routing tables automatically. It uses hop count as a metric, suitable for small networks.

Example Configuration:

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.10.0
Router(config-router)# network 192.168.20.0
Router(config-router)# network 192.168.30.0
```

DNS Server:

The Domain Name System (DNS) translates domain names into IP addresses, allowing users to access services using easy-to-remember names instead of numbers.

Example Configuration (Server in Packet Tracer):

- IP Address: 192.168.10.2
- Gateway: 192.168.10.1
- Add DNS record:
 - Name: www.campus.com
 - Address: 192.168.10.2

This enables clients to reach the web server by typing www.campus.com instead of its IP.

Project Development:

Required software:

SI No	Software Name	Version
01	Cisco packet Tracker	8.1.1

Designed topology:

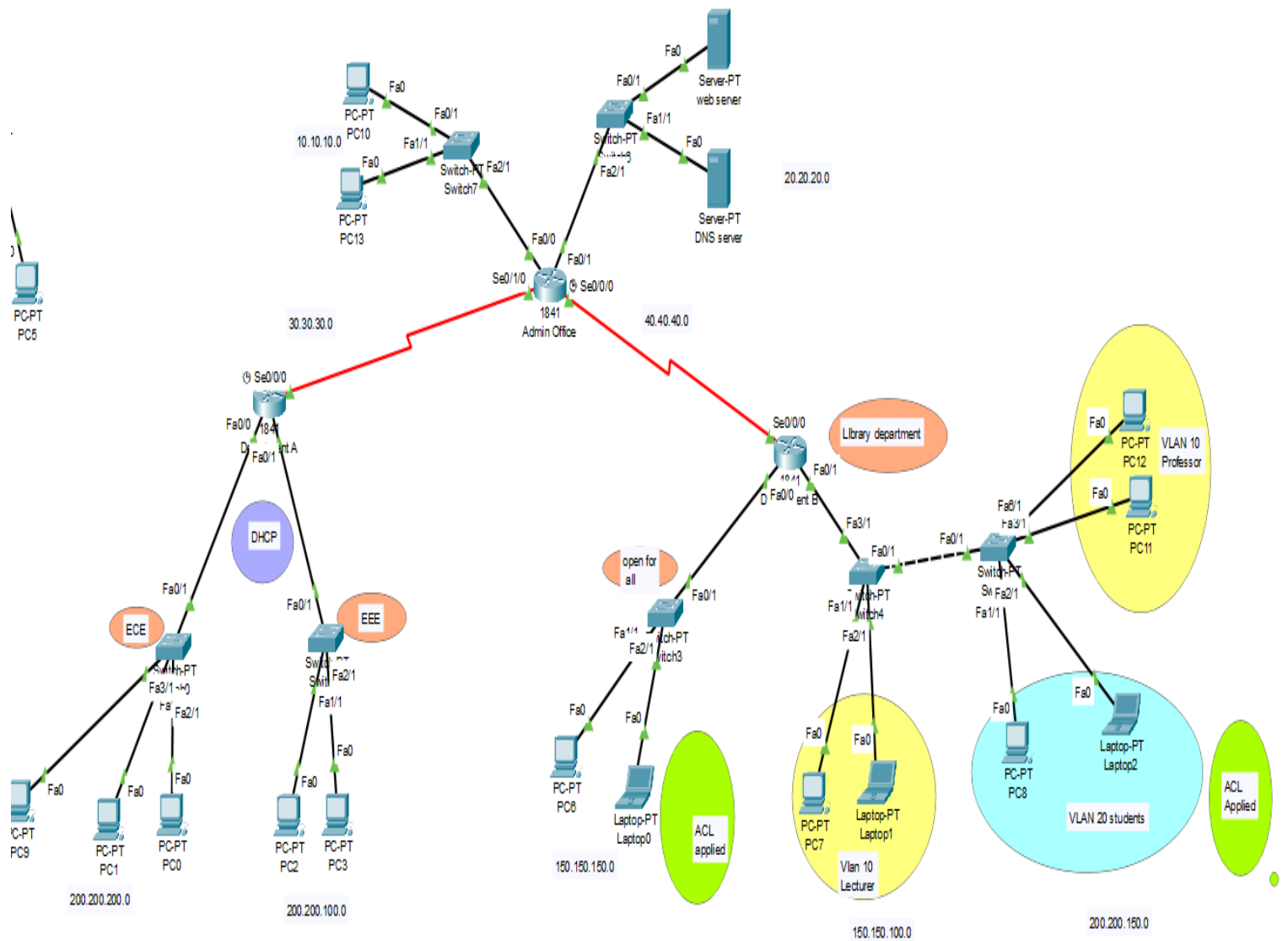


Fig 1: Topology of the campus network.

Design overview:

- I. Two departments (ECE & EEE)
- II. An admin office with DNS server and Web server.
- III. All password protected router and switch.
- IV. DHCP protocol for ECE and EEE.
- V. VLAN system for library department.
- VI. Dynamic routing protocol (RIP)
- VII. ACL service applied at the library section to limit users in admin office.

Configuration Process:

1. password protection:

- a. setting password for router:

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#line con 0
Router(config-line)#password cisco
Router(config-line)#log
Router(config-line)#login
Router(config-line)#login
```

Fig.2 password code for router at CLI

- b. setting password for server:

```
Switch#config tel
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#enable password class
```

fig.3 password for server at CLI.

2. DHCP protocol applying in ECE & EEE (R0):

```
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip d
Router(config)#ip dh
Router(config)#ip dhcp ex
Router(config)#ip dhcp excluded-address 200.200.200.1 200.200.200.10
Router(config)#ip dhc
Router(config)#ip dhcp poo
Router(config)#ip dhcp pool ECE
Router(dhcp-config)#network
Router(dhcp-config)#network 200.200.200.0 255.255.255.0
Router(dhcp-config)#defa
Router(dhcp-config)#default-router 200.200.200.1
Router(dhcp-config)#dn
Router(dhcp-config)#dns-server 20.20.20.2
Router(dhcp-config)#
```

Fig 4: DHCP Command at CLI of router (R0)

3.VLAN configuration (for Library department):

```
S1#configure terminal
Enter configuration commands, one per line.  End
S1(config)#vlan 10
S1(config-vlan)#name lecturer
S1(config-vlan)#exit
S1(config)#vlan 20
S1(config-vlan)#name students
S1(config-vlan)#exit
S1(config)#interface fa1/1
S1(config-if)#swic
S1(config-if)#swit
S1(config-if)#switchport mode ac
S1(config-if)#switchport mode access
S1(config-if)#switch acc
S1(config-if)#switch access vlan 10
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#
```

Fig 5: Vlan 10 configuration for faculty at S1

4. Creating sub-interface for VLAN networks:

```
library#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
library(config)#interface fa0/1.10
library(config-subif)#encap
library(config-subif)#encapsulation do
library(config-subif)#encapsulation dot1Q 10
library(config-subif)#ip address 150.150.100.1 255.255.255.0
library(config-subif)#interface fa0/1.20
library(config-subif)#encapsulation dot1Q 10

%Configuration of multiple subinterfaces of the same main
interface with the same VID (10) is not permitted.
This VID is already configured on FastEthernet0/1.10.

library(config-subif)#encapsulation dot1Q 20
library(config-subif)#ip address 200.200.150.1 255.255.255.0
library(config-subif)#exit
library(config)#
```

Fig 6: Creating sub-interface for router to connect with both vlan.

5. Applying dynamic routing protocol:

```
Router(config)#router rip
Router(config-router)#netwo
Router(config-router)#network 200.200.200.0
Router(config-router)#network 30.30.30.0
Router(config-router)#network 200.200.100.0
Router(config-router)#passi
Router(config-router)#passive-interface fa0/0
Router(config-router)#no auto
Router(config-router)#no auto-summary
Router(config-router)#
```

Fig 7: RIP implementation on all routers for dynamic routing

6. Implementation of ACL :

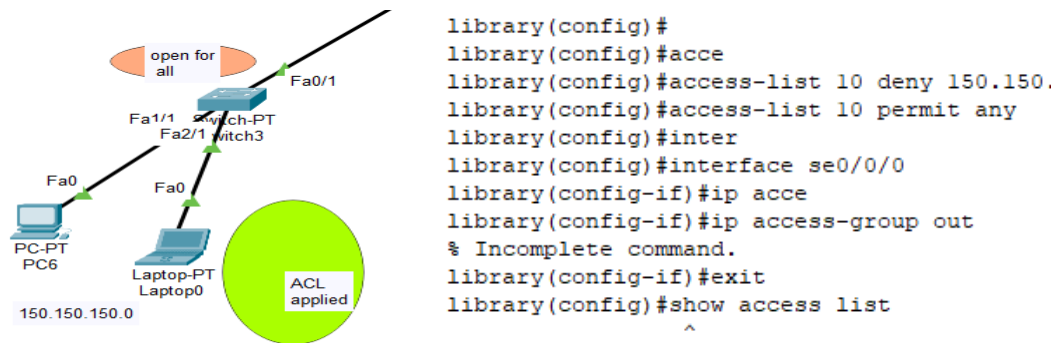


Fig 8: ACL implementation at “open for all” server to limit access in admin office

Result analysis:

1. Password encryption:

```
Switch>enable  
Password:  
Switch#
```

Fig 9: output of password encryption

2. DHCP functionality

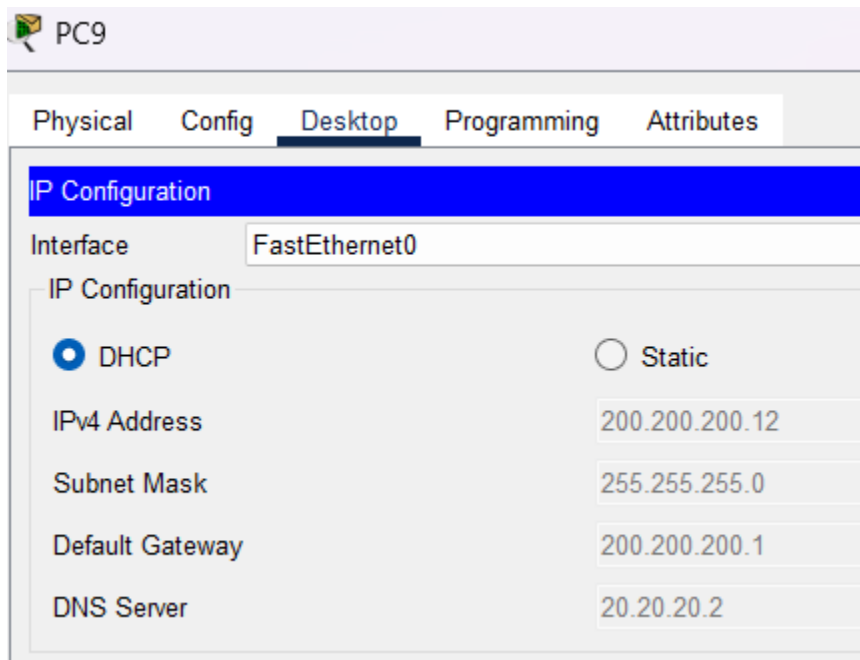


Fig 10 : output of dynamic host configuration

3. VLAN system verification:

Command :

show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa4/1, Fa5/1, Fa7/1
10	Professor	active	Fa3/1, Fa6/1
20	students	active	Fa1/1, Fa2/1
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

S2#

4. Fig11: output of Vlan brief

4. DNS server and Web server functionality:

Web Browser

< > URL Go

Hi Aurpa

Welcome to library account, Aurpa. Opening doors to new opportunities. Mind Wide Open.

Quick Links:

[A small page](#)
[Copyrights](#)
[Image page](#)
[Image](#)

Fig 12: output of webpage

5. Dynamic Routing Verification:

RIP Routing

Network

Network Address
30.0.0.0
200.200.100.0
200.200.200.0

Fig 13: output of RIP

6. ACL application:

	--	PC8	PC13	ICMP		0.000	N	0	(edit)	(delete)
	Failed	Laptop0	PC13	ICMP		0.000	N	1	(edit)	(delete)

Fig14: output of failed attempt to access admin office by “students”

Discussion:

In this project, a network system was designed to demonstrate the use of DHCP, VLANs, and ACLs for efficient communication and security management across different departments. The ECE and EEE departments were configured with DHCP servers under a common router to automatically assign IP addresses to devices, ensuring smooth and dynamic network connectivity without manual configuration. The Library department was divided into two VLANs — VLAN 10 for faculty and VLAN 20 for students — to logically separate their traffic within the same physical network. This separation ensured that students could not directly send data packets to the faculty network, maintaining data privacy and reducing broadcast traffic. A shared server was implemented to allow communication between both VLANs, but its access was carefully controlled through ACLs (Access Control Lists). The ACLs ensured that while both faculty and students could reach the shared server, their access to sensitive areas like the admin office was restricted. Additionally, the admin department contained a web server and a DNS server to support name resolution and web access for the entire network. However, only faculty members were permitted to access the admin section, while students were denied access for security reasons. This setup demonstrates how DHCP simplifies network configuration, VLANs improve segmentation and control, and ACLs strengthen network security by defining user privileges and communication boundaries.

Conclusion:

Overall, the project successfully integrated DHCP, VLAN, and ACL technologies to create a secure and well-structured departmental network. The system ensured efficient IP management, logical traffic separation, and controlled access between users and servers.

Reference:

1. Lab manual
2. [AI Infrastructure, Secure Networking, and Software Solutions - Cisco](#)
3. <https://www.geeksforgeeks.org/computer-networks>
4. <https://www.pcmag.com/how-to>
5. <https://www.fortinet.com>
6. <https://theconversation.com/what-is-dns-a-computer-engineer-explains>

